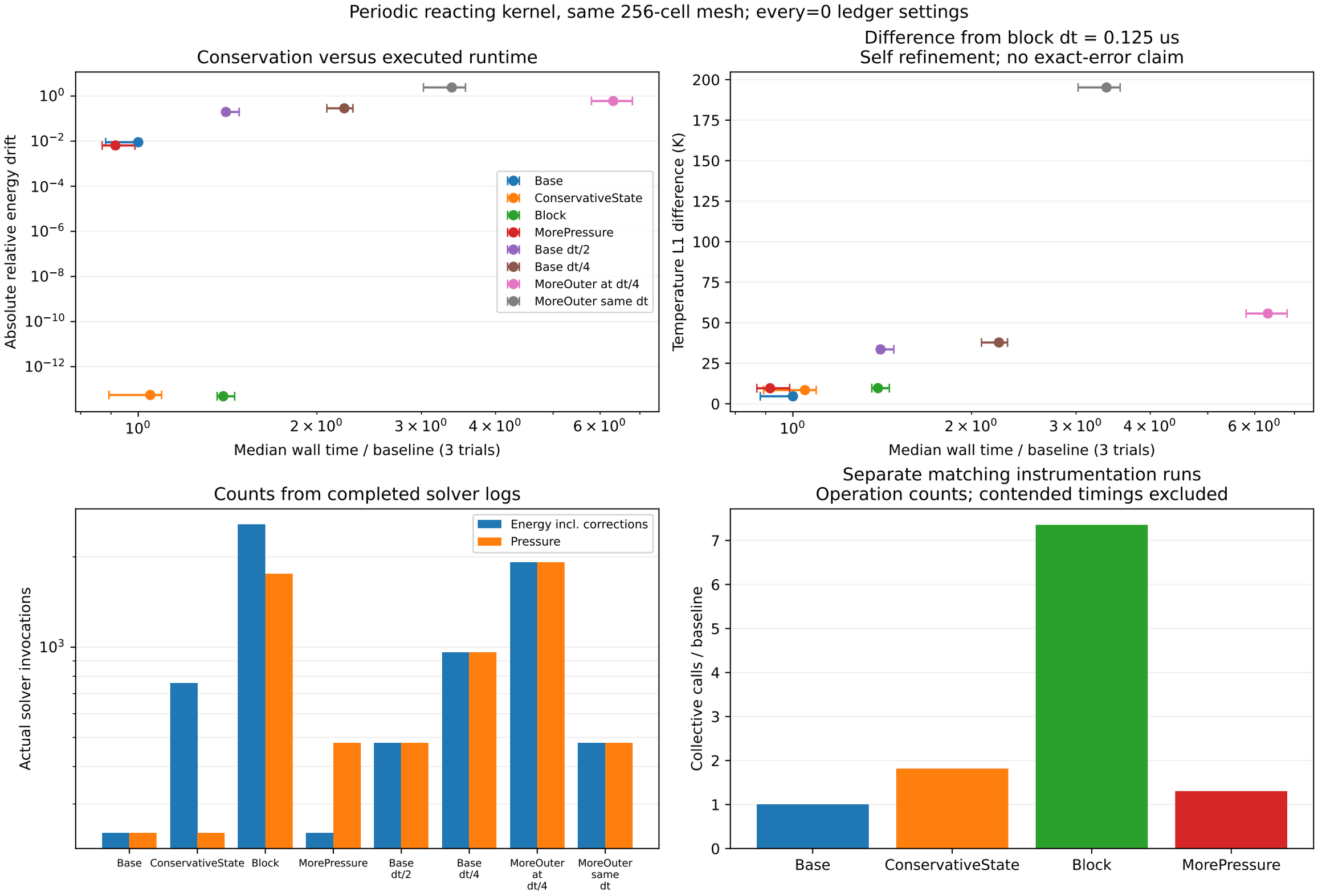
Periodic reacting kernel, same 256-cell mesh; every=0 ledger settings
Conservation versus executed runtime
Absolute relative energy drift
Median wall time / baseline (3 trials)
Base
ConservativeState
Block
MorePressure
Base dt/2
Base dt/4
MoreOuter at dt/4
MoreOuter same dt
Difference from block dt = 0.125 us
Self refinement; no exact-error claim
Temperature L1 difference (K)
Median wall time / baseline (3 trials)
Counts from completed solver logs
Actual solver invocations
Energy incl. corrections
Pressure
Base
ConservativeState
Block
MorePressure
Base dt/2
Base dt/4
MoreOuter at dt/4
MoreOuter same dt
Separate matching instrumentation runs
Operation counts; contended timings excluded
Collective calls / baseline
Base
ConservativeState
Block
MorePressure